Transformations & Vectors

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 2x + 9$ in solving equations with algebra, evaluate $f(5)$.

Final Answer: 19. Substitute $x = 5$: $f(5) = 2(5) + 9 = 19$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 3x + 3$ in methods for solving two-step equations, evaluate $f(2)$.

Final Answer: 9. Substitute $x = 2$: $f(2) = 3(2) + 3 = 9$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for methods for solving two-step equations consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 18.5. The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 4x + 4$ in two-step equations, evaluate $f(2)$.

Final Answer: 12. Substitute $x = 2$: $f(2) = 4(2) + 4 = 12$.

5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for two-step equations consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 19.5. The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 5x + 5$ in solving problems with equations, evaluate $f(2)$.

Final Answer: 15. Substitute $x = 2$: $f(2) = 5(2) + 5 = 15$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for solving problems with equations consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 20.5. The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

The curve $y = x^2 - 6x + 5$ can be mapped to $y = x^2$ by a sequence of two geometric transformations. Express the equation in vertex form, state the vertex, and describe fully each transformation.

Complete the square: $y = (x - 3)^2 - 9 + 5 = (x - 3)^2 - 4$. The vertex of the parabola is $(3, -4)$. To map $y = (x - 3)^2 - 4$ back to $y = x^2$, we must translate the graph 3 units to the left and 4 units upwards. In vector notation, this is a single translation by $(-3; 4)$, or two sequential transformations: (1) Translation by vector $(-3; 0)$ (3 units left), and (2) Translation by vector $(0; 4)$ (4 units up).

9 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Triangle ΔABC has vertices $A(1, 2)$, $B(3, 2)$, and $C(1, 5)$. The triangle is dilated by a scale factor of $k = 3$ centered at the origin, and then translated by vector $\langle -2, 4 \rangle$. What are the coordinates of vertex C''?

Step 1: Dilation centered at the origin: Multiply each coordinate of $C(1, 5)$ by the scale factor $k = 3$: $C(1, 5) \longrightarrow C'(3 \times 1, 3 \times 5) = C'(3, 15)$. Step 2: Translation by vector $\langle -2, 4 \rangle$: Add -2 to the x-coordinate and 4 to the y-coordinate: $C'(3, 15) \longrightarrow C''(3 - 2, 15 + 4) = C''(1, 19)$. The final coordinates of vertex C'' are $(1, 19)$.

10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Point $A(-2, 4)$ is reflected across the line $y = x$, and the resulting point is then translated by the vector $\langle 3, -1 \rangle$. What are the final coordinates?

Solution: Apply the transformations step by step: Step 1: Reflection across $y = x$ swaps the coordinates: $(x, y) \to (y, x)$: $A(-2, 4) \longrightarrow A'(4, -2)$. Step 2: Translation by $\langle 3, -1 \rangle$ adds 3 to the x-coordinate and subtracts 1 from the y-coordinate: $A'(4, -2) \longrightarrow A''(4 + 3, -2 - 1) = A''(7, -3)$. The final coordinates are $(7, -3)$.

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Find the inverse function $f^{-1}(x)$ for $f(x) = \frac{3x+4}{2x-5}$, and state the domain of f^{-1}.

Let $y = \frac{3x+4}{2x-5}$. Swap variables: $x = \frac{3y+4}{2y-5}$. Multiply both sides by $2y - 5$: $x(2y - 5) = 3y + 4 \Rightarrow 2xy - 5x = 3y + 4$. Group all y-terms on one side: $2xy - 3y = 5x + 4 \Rightarrow y(2x - 3) = 5x + 4$. Divide to isolate y: $f^{-1}(x) = \frac{5x+4}{2x-3}$. Domain of f^{-1}: $x \neq \frac{3}{2}$ (all real numbers except $x = 1.5$).

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12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Given $f(x) = \frac{2}{x-1}$ and $g(x) = \sqrt{x+5}$, find (a) the value of $(f \circ g)(4)$, and (b) the domain of $g(x)$.

1. Evaluate the inner function first: $g(4) = \sqrt{4+5} = \sqrt{9} = 3$. 2. Substitute into f: $(f \circ g)(4) = f(g(4)) = f(3) = \frac{2}{3-1} = \frac{2}{2} = 1$. 3. Domain of $g(x) = \sqrt{x+5}$: The radicand must be non-negative: $x + 5 \geq 0 \Rightarrow x \geq -5$ or in interval notation $[-5, \infty)$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The curve $y = f(x)$ has a stationary point at $(4, -6)$. State the coordinates of the corresponding stationary point on the curve: (a) $y = f(x+3) + 2$, (b) $y = 3f(2x)$, (c) $y = -f(-x)$.

(a) $y = f(x+3) + 2$ translates 3 units left and 2 units up: $(4-3, -6+2) = (1, -4)$. (b) $y = 3f(2x)$ stretches horizontally by $\frac{1}{2}$ and vertically by 3: $\left(4 \times \frac{1}{2}, -6 \times 3\right) = (2, -18)$. (c) $y = -f(-x)$ reflects in both axes (point reflection through the origin): $(-4, -(-6)) = (-4, 6)$.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

[Functions & Transformations · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 3.01 Coordinate Geometry**

[Transformations & Symmetry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 4.01 Mensuration**

[Graphs & Transformations · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 5.01 Geometry**

[Vectors in Two Dimensions & Relative Motion · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

Substitute $t = 4$: $g(4) = 3(4) + 3 = \mathbf{15}$. Setting $3t + 3 = 18$ gives $3t = 15 \Rightarrow t = \mathbf{5}$.

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18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Trigonometry

A car of mass 1200 kg ascends a hill of inclination α, where $\sin\alpha = \frac{1}{20}$, against a constant resistance to motion of 500 N. Taking $g = 9.8$ m/s^2: (a) find the driving force required to travel at a steady speed of 25 m/s, (b) calculate the power developed by the engine in kilowatts.

(a) Component of weight down the slope: $W_{\parallel} = mg\sin\alpha = 1200(9.8)\left(\frac{1}{20}\right) = 60(9.8) = 588$ N Since the car travels at steady speed (acceleration $a = 0$), the engine driving force balances all resistance:

$F_{\text{drive}} = \text{Resistance} + mg\sin\alpha = 500 + 588 = 1088$ N (b) Power output:

$P = F_{\text{drive}} \times v = 1088 \times 25 = 27\,200$ W $= 27.2$ kW

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.01 Probability & Statistics

Events A and B are such that $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cup B) = 0.8$. Find $P(A \cap B)$, calculate $P(A|B)$, and determine whether A and B are independent.

1. Use the addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B) \Rightarrow 0.8 = 0.6 + 0.5 - P(A \cap B) \Rightarrow P(A \cap B) = 1.1 - 0.8 = 0.3$. 2. Conditional probability: $P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.3}{0.5} = 0.6$ 3. Check for independence: $P(A \cap B) = 0.3$ and $P(A) \times P(B) = 0.6 \times 0.5 = 0.30$. Since $P(A \cap B) = P(A)P(B)$ (or equivalently $P(A|B) = P(A) = 0.6$), events A and B are independent.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 9.01 Functions

Find the sum of all integers between 10 and 100 (inclusive) that are divisible by 4.

1. Identify the sequence: The first multiple of $4 \geq 10$ is $a_1 = 12$. The last multiple ≤ 100 is $a_n = 100$. The common difference is $d = 4$. 2. Find the number of terms n:

$a_n = a_1 + (n-1)d \Rightarrow 100 = 12 + (n-1)4 \Rightarrow 88 = 4(n-1) \Rightarrow n - 1 = 22 \Rightarrow n = 23.$ 3. Calculate the sum using $S_n = \frac{n}{2}(a_1 + a_n)$:

$S_{23} = \frac{23}{2}(12 + 100) = \frac{23}{2}(112) = 23 \cdot 56 = 1288.$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 5x + 7$ in introduction to equations, evaluate $f(2)$.

a. 17 ✓

b. 22

c. 10

d. 21

Substitute $x = 2$: $f(2) = 5(2) + 7 = 17$.

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 6x + 8$ in solutions to equations, evaluate $f(2)$.

a. 20 ✓

b. 26

c. 12

d. 24

Substitute $x = 2$: $f(2) = 6(2) + 8 = 20$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for introduction to equations consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5 ✓

d. 26.5

The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for solutions to equations consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5 ✓

d. 18.5

The sum is $10 + 14 + 18 + 24 = 66$, and dividing by 4 gives 16.5.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The diagram shows point A, point B, and their images A' and B' after a transformation. The transformation can be described by the vector $(p; q)$. Find the value of p.

a. **3** ✓

b. -3

c. 5

d. -5

The transformation maps point A $(1, 2)$ to point A' $(-2, 5)$. A translation is represented by a vector $(x' - x; y' - y)$, where (x, y) is the original point and (x', y') is its image. Therefore, the translation vector is $(-2 - 1; 5 - 2) = (-3; 3)$. Comparing this to the given vector $(p; q)$, we have $p = -3$ and $q = 3$. However, let's re-examine the diagram. Point A is at $(-4, 2)$ and A' is at $(-1, 2)$. The vector would be.

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)
Point $P(-2, 4)$ is translated by the vector $(2; 1)$. Find the image of P.

a. $(0, 5)$ ✓

b. $(-4, 3)$

c. $(0, 4)$

d. $(-2, 5)$

Applying Transformations & Vectors principles, Image $= (-2 + 2, 4 + 1) = (0, 5)$. Therefore, the verified correct option is $(0, 5)$.

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)
Rotate the point $(1, 7)$ by 180° about the origin. Find the image.

a. $(1, -7)$

b. $(-1, -7)$ ✓

c. $(1, 7)$

d. $(-1, 7)$

Rotation by 180° about the origin maps (x, y) to the image $(-1, -7)$.

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)
Rotate the point $(3, -5)$ by 270° anticlockwise about the origin. Find the image.

a. $(-5, -3)$ ✓

b. $(3, 5)$

c. $(-3, -5)$

d. $(3, -5)$

Rotation by 270° about the origin maps (x, y) to the image $(-5, -3)$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)
Rotate the point $(4, 5)$ by 90° clockwise about the origin. Find the image.

a. $(-4, 5)$

b. $(5, -4)$ ✓

c. $(4, 5)$

d. $(4, -5)$

Rotation by -90° about the origin maps (x, y) to the image $(5, -4)$.

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Reflect the point $(-1, 2)$ in the line $y = x$. Find the image.

a. $(2, 1)$

b. $(-1, 2)$

c. $(1, -2)$

d. $(2, -1)$ ✓

Reflection in the line $y = x$ maps (x, y) appropriately: image $= (2, -1)$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Reflect the point $(9, 1)$ in the x-axis. Find the image.

a. $(1, -9)$

b. $(-9, -1)$

c. $(9, 1)$

d. $(9, -1)$ ✓

Reflection in the x-axis maps (x, y) appropriately: image $= (9, -1)$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The graph of $y = f(x)$ is shown. Describe the transformations to get $y = f(x) + 2$ and $y = f(-x)$.

a. **Translation up 2; reflection in y-axis** ✓

b. Translation up 2; reflection in x-axis

c. Translation right 2; reflection in y-axis

d. No transformation

$f(x) + 2$: translate 2 up. $f(-x)$: reflect in y-axis.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the magnitude of the vector $(5; 12)$.

a. **13** ✓

b. 24

c. 22

d. 16

$|\vec{v}| = \sqrt{(5)^2 + (12)^2} = \sqrt{169} = 13$

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Point $(6, 5)$ is enlarged by scale factor 2 about the origin. Find the image.

a. $(10, 8)$

b. $(14, 10)$

c. $(12, 10)$ ✓

d. $(12, 12)$

Image $= (center + k(point - center)) = (12, 10)$.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Point $(6, 5)$ is enlarged by scale factor 3 about $(3, 2)$. Find the image.

a. $(10, 9)$

b. $(12, 11)$ ✓

c. $(12, 13)$

d. $(14, 11)$

Image $= (center + k(point - center)) = (12, 11)$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the median of the data set: $1, 2, 6, 13, 14$.

a. 9

b. 13

c. **6** ✓

d. 7

With 5 values sorted, the median is the middle (3th) value: 6.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the mean of the data set: $5, 6, 6, 16, 17$.

a. **10** ✓

b. 14

c. 9

d. 0

Mean $= \frac{5 + 6 + 6 + 16 + 17}{5} = \frac{50}{5} = 10$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Trigonometry

The function f is defined by $f(x) = 2x^2 - 3x + 1$. Find the value of $f(-3)$.

a. 28 ✓

b. 10

c. -8

d. -26

To find $f(-3)$, substitute $x = -3$ into the function definition:

$f(-3), = 2(-3)^2 - 3(-3) + 1; [4pt], = 2(9) - (-9) + 1; [4pt], = 18 + 9 + 1; [4pt], = 28$ Common errors: - Evaluating $-3(-3)$ as -9 yields $18 - 9 + 1 = 10$. - Squaring -3 incorrectly as -9 yields $2(-9) + 9 + 1 = -8$. - Making both sign errors yields $-18 - 9 + 1 = -26$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.01 Probability & Statistics

The probability that it rains on Monday is 0.3. If it rains on Monday, the probability it rains on Tuesday is 0.6. If it does not rain on Monday, the probability it rains on Tuesday is 0.2. What is the probability that it rains on at least one of the two days?

a. **0.44** ✓

b. 0.56

c. 0.50

d. 0.62

Calculate the complement $P(\text{no rain on either day})$: $P(\text{no rain Mon}) = 1 - 0.3 = 0.7$. $P(\text{no rain Tue} \mid \text{no rain Mon}) = 1 - 0.2 = 0.8$. $P(\text{no rain either day}) = 0.7 \times 0.8 = 0.56$. Therefore, $P(\text{rain at least once}) = 1 - 0.56 = 0.44$. [M1 for 0.7 * 0.8 = 0.56, A1 for 1 - 0.56 = 0.44]

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 9.01 Functions

On a histogram, what does the area of each bar represent?

a. The class width

b. **The frequency** ✓

c. The frequency density

d. The cumulative frequency

In a histogram, the area of each rectangular bar is proportional to (or equal to) the frequency of that class: $\text{Frequency} = \text{Frequency density} \times \text{Class width}$. [B1]

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Number

Convert the recurring decimal $0.\dot{2}\dot{7}$ into a fraction in its simplest form.

a. $\frac{27}{100}$

b. $\frac{3}{11}$ ✓

c. $\frac{9}{33}$

d. $\frac{1}{4}$

Let $x = 0.272727\ldots$. Then $100x = 27.272727\ldots$. Subtract: $100x - x = 27 \Rightarrow 99x = 27 \Rightarrow x = \frac{27}{99} = \frac{3}{11}$. [M1 for subtraction method, A1 for 3/11]

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 10.01 Sets & Venn Diagrams

The length of a rectangle is 12 cm correct to the nearest millimetre, and its width is 8 cm correct to the nearest millimetre. Calculate the upper bound for the area of the rectangle in cm^2.

a. **(97.0025)** ✓

b. (96.0000)

c. (95.0025)

d. (98.0125)

The upper bound for length is $12 + 0.05 = 12.05$ cm and for width is $8 + 0.05 = 8.05$ cm, so the maximum area is $12.05 \times 8.05 = 97.0025\ \text{cm}^2$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Algebra & Graphs

A cumulative frequency graph for 120 marathon runners shows that the 90th percentile finish time is 3 hours 45 minutes. How many runners finished in a time faster than 3 hours 45 minutes?

a. 12 runners

b. 90 runners

c. **108 runners** ✓

d. 100 runners

The 90th percentile corresponds to 90% of 120 runners: $0.90 \times 120 = 108$ runners. These 108 runners finished at or below this time (faster). [M1 for 0.90*120, A1 for 108]

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Coordinate Geometry

Rationalise the denominator of $\frac{6}{\sqrt{3}}$.

a. $2\sqrt{3}$ ✓

b. $3\sqrt{3}$

c. $\sqrt{2}$

d. $6\sqrt{3}$

Multiply numerator and denominator by $\sqrt{3}$: $\frac{6\sqrt{3}}{\sqrt{3}\sqrt{3}} = \frac{6\sqrt{3}}{3} = 2\sqrt{3}$. [M1 for multiplying by sqrt(3)/sqrt(3), A1 for 2sqrt(3)]

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Mensuration

Work out $\frac{3}{5} \div \frac{9}{10}$. Give your answer as a fraction in its simplest form.

a. $\frac{2}{3}$ ✓

b. $\frac{27}{50}$

c. $\frac{3}{2}$

d. $\frac{1}{3}$

Multiply by the reciprocal: $\frac{3}{5} \times \frac{10}{9} = \frac{3 \times 10}{5 \times 9} = \frac{30}{45} = \frac{2}{3}$. [M1 for multiplying by reciprocal, A1 for 2/3]

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